

Department of Computer Science

Federal Urdu University

of Arts, Science and Technology

ARTIFICIAL INTELLIGENCE

ASSIGNMENT

Topic: Smart ISP Recommendation Agent

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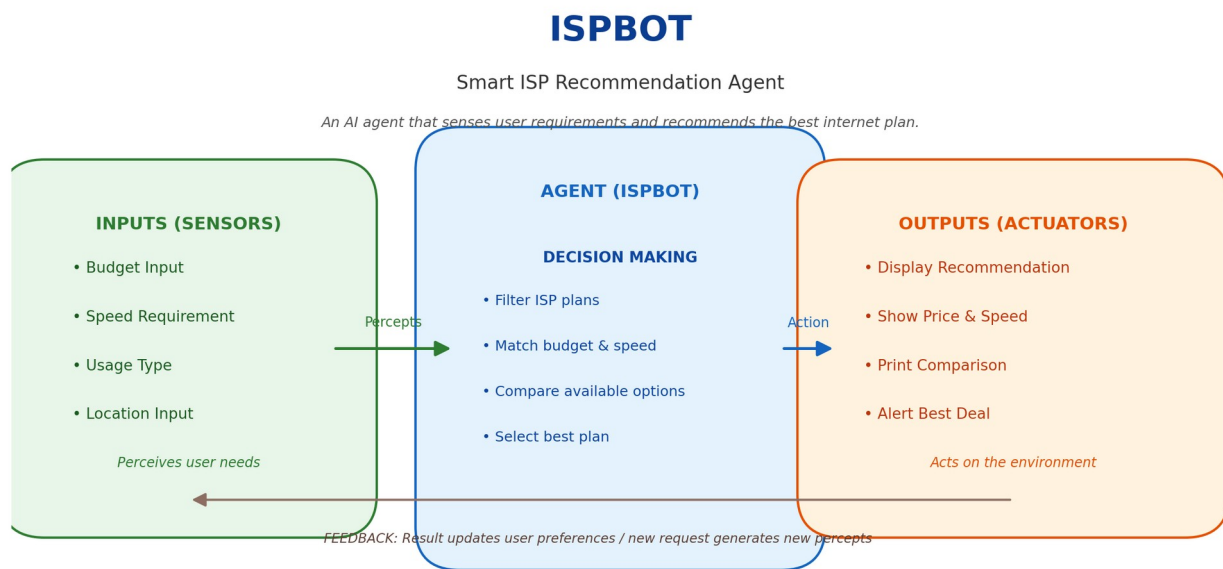
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ISPBOT

ISPBOT is an intelligent recommendation agent that helps users choose the best internet service provider (ISP) plan. It takes the user's budget, required internet speed, and intended usage (browsing, streaming, or gaming) as input, then searches through a database of available ISP plans and recommends the most suitable option.

Graphical Representation

The working of ISPBOT can be shown as a simple sense-think-act cycle. The process starts when the agent receives the user's requirements as percepts, then reasons over the available ISP plans to decide on the best match, and finally acts by displaying the recommendation to the user.



Agent Formulation

The ISPBOT agent is an entity that receives the user's requirements as input, reasons over a database of ISP plans, and produces a recommendation as output.

Environment

The environment consists of the ISP market: the list of available providers, their prices, internet speeds, and the type of usage each plan is best suited for, along with the requirements entered by the user.

Sensors (Inputs)

- **Budget Input:** The maximum amount the user is willing to pay per month.
- **Speed Requirement:** The minimum internet speed (Mbps) the user needs.
- **Usage Type:** Whether the connection is mainly for browsing, streaming, or gaming.

- Location Input: The user's area, used to check which ISPs are available there.

Actuators (Outputs)

- Display Recommendation: Shows the name of the recommended ISP provider.
- Show Price & Speed: Displays the plan's monthly cost and speed.
- Print Comparison: Optionally lists all suitable plans side by side.
- Alert Best Deal: Notifies the user if an especially good plan is found.

Types of Environment

Property	Type	Justification
Observability	Fully Observable	The agent has complete access to ISP plan data (price, speed, type) needed to make a decision.
Determinism	Deterministic	Given the same user inputs, the agent always produces the same recommendation.
Episodes	Episodic	Each recommendation request is independent of previous requests.
Change	Static	The ISP plan database does not change while the agent is making a decision.
Values	Discrete	Inputs and outputs (plan names, prices, speeds) are countable, distinct values.
Agents	Single-Agent	Only one agent (ISPBOT) operates in this environment; there is no competition or cooperation with other agents.

PEAS Framework

Component	Description
P – Performance	Accurate recommendation, cost-effectiveness, user satisfaction, fast response
E – Environment	ISP plans database, user requirements, budget, market prices
A – Actuators	Display screen output, comparison table, alert/notification
S – Sensors	Budget input, speed requirement input, usage type input, location input